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Started	Wednesday, 21 October 2026, 8:55 PM
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Duration	3 mins 16 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Hash

b. Binary ✓

c. Linear

d. Both (1) & (3)

Question **2**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list

b. Binary search has a lower time complexity on large, sorted lists ✓

c. Binary search can find multiple instances of the target element

d. Binary search works on unsorted lists

Question **3**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎

b. O(n log n)

c. O(1) ✓

d. O(log n)

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By comparing each element in the list sequentially ✓
c. By dividing the list into halves
d. By using a hash function

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Merge search
b. Bubble search
c. Linear search ✓
d. Quick search

Question **6**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It works on unsorted lists
c. It can be applied only if the list is sorted ✓
d. It always starts from the beginning of the list

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The search stops
c. The search continues in the left or right sublist ✓
d. The list is sorted

Question **8**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 99 ✓
- b. 89 and 94
- c. 90 and 94
- d. 89 and 99

Question **9**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 89 and 94
- b. 90 and 100
- c. 90 and 99 ✓
- d. 94 and 99

Question **10**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the element is found in the list ✓
- b. When the search algorithm finishes
- c. When the list contains duplicate elements
- d. When the list is sorted

Question **11**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Bubble ✓

c. Insertion

d. Selection

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓

b. When the list is very small

c. When the list contains duplicate elements

d. When the list is unsorted

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly

b. Sorting the list before searching

c. Skipping every second element

d. Checking each element sequentially ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements

b. The search continues until the end of the list

c. The search starts over from the beginning

d. The search stops immediately ✓

Question **15**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 10 ✓
c. 1000
d. 500